

Stage 1: Multi-Source Literature Retrieval and Deduplication

We retrieve papers from three complementary academic databases to maximize coverage and enable cross-source deduplication:

arXiv. We query the arXiv Atom API using phrase-matched searches including `all:"active inference"`, `all:"free energy principle"`, `all:"expected free energy"`, `all:"variational free energy" AND all:"inference"`, and targeted Energy-Based Model queries (`all:"energy-based model" AND all:"free energy"`, `all:"Helmholtz machine" AND all:"inference"`, `all:"Boltzmann machine" AND all:"free energy"`, `all:"contrastive divergence" AND all:"generative model"`). The `all:` prefix searches titles, abstracts, and full text; phrase matching reduces contamination from unrelated physics papers that mention “free energy” in thermodynamic contexts. The EBM-adjacent queries capture research at the intersection of energy-based generative modeling and variational inference—a growing convergence area